

Coordination of Mood Phrases in Korean*

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Jae-Il Yeom. 2019. Coordination of Mood Phrases in Korean. *Language and Information* 23.2, 59-96. In this paper, I attempt to find the answers to the questions of what a speech act comes from and how a MP is interpreted. To do this, I discuss whether Mood Phrases (= MPs) can be conjoined or disjoined, and why they can or cannot, in matrix and embedded clauses. If a MP is used as a matrix one, it becomes a speech act. If some phenomena are observed only in matrix MPs, they are phenomena of speech acts. If some phenomena are observed in embedded MPs, they are phenomena of MPs. Based on these criteria, I claim that a speech act comes from the utterance of a MP as a separate sentence, not from the MP itself. This is also supported by the fact that there is no conjunction connective that combines two MPs. Based on these observations, I also claim that *kuliko* 'and' and *hokun* 'or' are coordinators of speech acts. Speech acts can be conjoined relatively freely, but they can be disjoined only if a preceding speech act can be re-interpreted as contributing to a larger speech act. And there is only one coordinator *-kena* 'or' that disjoins embedded MPs and there is no conjunction connective of embedded MPs. I explain all the observations based on the claim that the use of a MP presupposes an event or state that involves the meaning of the MP. It is an utterance event when a matrix MP is used and it is an event or state introduced in the context when an embedded MP is used. This allows us to explain that MPs can be disjoined only if the disjunction connective takes wide scope over a predicate that introduces an event or state that involves the MPs, and leads to the meaning of conjunction. Hence no conjunction connective of MPs is necessary. (Hongik University)

Key words: coordination, mood phrases, disjunction, conjunction, speech act

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1. Introduction

In English, there are phenomena observed only in root clauses, as pointed out in Emonds (1969) and Ross (1970). Neither Left Dislocation nor subject-aux inversion is observed in embedded clauses, and there is no embedded imperative:

- (1) a. This room, it really depresses me.
b. *They put so much furniture in here that [this room, it really depresses me].
- (2) a. What do you want?
b. *I do not know what do you want.
- (3) a. Don't go!
b. *I told you don't go.

However, even if a form of a root clause is not freely embedded, there are phenomena that show speech acts are embedded: quantification into questions, conditional speech acts, etc. There are also other cases where an embedded clause has a form of a root clause. In such a case the embedded clause can be taken to be a speech act:

- (4) a. Mary thinks/said John has left. (English)
b. Maria glaubt/sagt, Johann ist abgereist. (German)

Hooper & Thompson (1973) claim that the special status of a root clause comes from the fact that a root clause has a function of a speech act. Thus the embedded clauses here carry the main points of the entire sentences. By the utterances, the speakers actually assert the propositions the embedded clauses denote. See Hopper & Thompson (1973), Heycock (2006) and Krifka (2014) for more discussions.

This seems to be a semantic and pragmatic proposal about speech acts. In this type of analysis, it is assumed that there are forms of root clauses and that they are only exceptionally used in embedded clauses when the speech acts of the matrix sentences are determined by the embedded clauses. In Korean, however, there are mood markers which can occur in both root clauses and embedded clauses and head Mood Phrases (= MPs): *-ta* is a declarative mood marker, *-nya* an interrogative, *-(e|u)la* an imperative, *-ca* a hortative, and *-(u)ma* a promissive:¹⁾

1) More precisely, *-(e)la* is only used as a default imperative mood marker in matrix clauses and *-(u)la* can be used in both matrix and embedded clauses. There are other mood markers used only in matrix clauses: *-(e)* and *-o* as declarative, imperative or interrogative mood markers depending on the intonation, *-(u)pni-kka* 'hon.dec-int' and *-(u)l-kka* 'mod-int' as interrogative mood markers with some

- (5) a. hayngpokha-{ta., nya?, y(e)la!, ca!, ma.
 happy-{dec, int, imp, hor, prm}²⁾
 ‘{pro is happy., Is pro happy?, Be happy!, Let’s be happy!, I promise to be happy.}
- b. hayngpokha-{ta-ko malha, nya-ko mwul, la-ko malha, ca-ko malha,
 happy-{dec-cmp say, int-cmp ask, imp-cmp say, hor-cmp say,
 ma-ko yaksokha}-yess-ta.³⁾
 prm-cmp promise}-pst-dec
 ‘{pro said pro was happy, pro asked if pro was happy, pro said to be happy, pro
 promised to be happy}’

This indicates that it is not exceptional that forms of matrix clauses can occur in embedded clauses. From this, we can ask a lot of questions. We can start with the question of whether a MP denotes a speech act. If it does, then an embedded clause can denote a speech act. If it does not, then a root clause will not denote a speech act.

On the other hand, Krifka (2001) mentions restrictions on coordination of matrix clauses under the same assumption: speech acts are conjoined but not disjoined.

- (6) A: Which dish did Al make? And which dish did Bill make?
 B: Al (made) the pasta, and Bill the salad.
- (7) A: Which dish did Al make? Or which dish did Bill make?

(6) is fine, but (7) is odd. If (7) is to be fine, the second question is understood as canceling the first sentence, or replacing the first with the second, not as a disjunction of two speech acts. Interestingly, similar restrictions are observed in Korean:

- (6’) inho-nun eti ka-ss-nya? kuliko mina-nun eti ka-ss-nya?
 Inho-top where go-pst-int and Mina-top where go-pst-int
 ‘Where did Inho go? And where did Mina go?’

restrictions, *-(u)psi-ta* ‘hon.hor-dec’ as a hortative mood marker, *-(u)lkkey* as a promissive mood marker, etc.

- 2) In this paper, I use the following abbreviations of grammatical terms: acc(usative case), adn(ominal ending), cl(assifier), c(o)mp(lementizer), dec(larative mood), hon(orific marker), hor(tative mood), imp(er)f(ective aspect), int(errogative mood), mod(al operator), n(or)m(ina)l(izer), nom(inative case), p(a)st, p(e)rf(ect), pl(ural), pr(o)g(ressive aspect), pr(o)m(issive mood), top(ic), etc.
- 3) One reviewer points out that the expression *-ma-ko* is not so natural as the other combinations. That form is used less and less to the extent that some native speakers hesitate to accept it. But we can find a lot of examples of its use in texts written by native speakers of Korean on the Internet.

- (8) *inho-nun eti ka-ss-nya? hokun mina-nun eti ka-ss-nya?
 Inho-top where go-pst-int or Mina-top where go-pst-int
 'Where did Inho go? Or, where did Mina go?'

This implies that in Korean a root clause may denote a speech act. But then an embedded clause also can denote a speech act, which we do not want to conclude. This is related to the issue of whether a mood marker gives the meaning of a speech act.

In this paper, I will discuss coordinations of speech acts and MPs in Korean in relation to the issues of whether a speech act can be embedded, what a mood marker means, and why MPs are not easily disjoined, and provide answers to them. In doing this, I discuss imperatives, hortatives, or promissives sometimes, but I mainly discuss declarative and interrogative sentences, because the semantics of other sentences are not well-established. From these discussions, I will claim that a MP is interpreted with respect to a situation in which it is used. A matrix MP is interpreted with respect to the utterance context.

The paper is organized as follows. In Section 2, I discuss basic observations to consider in the paper in relation to coordination of speech acts or MPs. In Section 3, I discuss coordination of speech acts. *Kuliko* 'and' and *hokun* 'or' combine speech acts. For this reason, they are attached to one MP, but they do not combine MPs. Conjunction of speech acts is allowed, but disjunction of speech acts is allowed only when a preceding speech act can be re-interpreted as an incomplete speech act. In Section 4, I discuss coordination of embedded MPs. First I discuss properties of embedded MPs. Then I claim that a MP is used with respect to a single event or state. This results in the restriction that no MPs are conjoined and that MPs are disjoined only if the disjunction connective has wide scope over a predicate that introduces an event or state to which each of the MPs is related. In Section 5, I explain why no conjunction of MPs arises, and why a speech-act adverbial occurs with a matrix MP, based on the requirement of an event or state in relation to the use of a MP. In Section 6, I conclude the paper.

2. Coordination facts of speech acts & MPs

2.1 Conjunction of speech acts

In this section, I will try to show what connectives combines matrix MPs and what restrictions they show. For a spatial reason, I will discuss only frequently used ones. In

Korean, there are various conjunction and disjunction connectives. In dealing with various connectives, it is necessary to discuss what conditions must be met when a connective can be considered as a coordinator, and what conditions must be met by the elements connected by a coordinating connective. the former conditions, see Kwon (1985), Ywu (1986), Kim (1988), among others, and for the latter, see Lee & Im (1983), Koh (2003), etc. Here I will consider typical examples, which satisfy the conditions.

In this paper, I am more interested in the restrictions on coordinations of MPs in Korean, but I have not found any studies on them. First, I will start with what connectives can conjoin MPs. There are three main conjunction connectives:

- (9) a. *-(k)wa*: Ns/NPs (without a case marker), PPs (nominal conjunction connective)
- i. inho-wa mina 'Inho and Mina'
 - ii. kang-eyse-wa pata-eyse 'in the river and in the sea'
 - river-in-and sea-in
 - iii. *inho-ka wul-ess-ta-wa mina-nun wus-ess-ta.
 - Inho-nom cry-pst-dec-and Mina-top laugh-pst-dec
 - 'Inho cried. And Mina laughed.'
- b. *-ko*: Vs/VPs, AspPs and TPs (verbal conjunction connective)
- i. wul-ko wus-nun-ta. '(be) crying and laughing'
 - cry-and laugh-impf-dec
 - ii. inho-ka wul-{ko.iss, ess}-ko mina-ka wus-{ko.iss, ess}-ta.⁴⁾

4) If smaller phrases than MPs are conjoined, they generally share the same mood. There are exceptions to this generalization. In the following example, the first TP is not in the scope of the interrogative mood marker:

- I. na-nun sakwa-lul mek-ess-ko, ne-nun mwues-ul mek-ess-nya?
- I-top apple-acc eat-pst-and you-top what-acc eat-pst-int
- 'I ate the apple and what did you eat?'

(i) is fine and the interrogative mood marker only has scope over the second conjunct. Otherwise, the question word *mwues* 'what' would violate the Coordinate Structure Constraint as in the following:

- ii. *ne-nun mwues-ul mek-ess-ko, na-nun sakwa-lul mek-ess-nya?
- you-top what-acc eat-pst-and I-top apple-acc eat-pst-int
- 'What did you eat, and I ate an apple?'

An example like (i) is allowed when the first conjunct conveys what is given in the preceding context:

- iii. a. A: sakwa-wa pay-ka iss-nuntey ettehkey ha-lkka?
- apple-and pear-nom be-and how do-mod.int

- Inho-nom cry-{prg, pst}-and Mina-nom laugh-{prg, pst}-dec
 'Inho {was crying, cried} and Mina {was laughing, laughed}.'
- iii. *inho-ka wul-ess-ta-ko mina-nun wus-ess-ta.
 Inho-nom cry-pst-dec-and Mina-top laugh-pst-dec
 'Inho cried and Mina laughed.'
- c. *kuliko*: NPs (with or without a case marker), PPs, AdvPs, MPs, CPs
- i. na-nun inho(-lul), kuliko mina-lul manna-ss-ta.
 I-top Inho(-acc), and Mina-acc meet-pst-dec
 'I met Inho and Mina.'
- ii. kang-eyse kuliko pata-eyse 'in the river and in the sea'
 river-in and sea-in
- iii. na-nun cha-lul coyonghi, kuliko chenchhenhi masi-ess-ta.
 I-top tea-acc quietly and slowly drink-pst-dec
 'I drank tea quietly and slowly.'
- iv. inho-ka wul-ess-ta. kuliko mina-nun wus-ess-ta.
 Inho-nom cry-pst-dec. and Mina-top laugh-pst-dec
 'Inho cried. And Mina laughed.'
- v. mina-ka o-ass-ta-ko, kuliko pro ca-n-ta-ko ha-yess-ta
 Mina-nom come-pst-dec-cmp and sleep-npst-dec-cmp say-pst-dec
 '(He) said that Mina came and that she was sleeping.'

From these examples, it is clear that only *kuliko* is attached to a MP. One morphological property is that it is a free morpheme, while others are bound ones.

'There's an apple and a pear. What shall we do (with them)?'

- b. B: ??ney-ka sakwa-lul mek-ko nay-ka pay-lul mek-uma.
 you-nom apple-acc eat-and I-nom pear-acc eat-prm
 'You eat the apple, and I promise to eat the pear.'

- c. B: nay-ka sakwa-lul mek-ul.kkey.
 I-nom apple-acc eat-mod.dec
 'I will have the apple.'

- A: kulemyen ney-ka sakwa-lul mek-ko nay-ka pay-lul mek-uma.
 then you-nom apple-acc eat-and I-nom pear-acc eat-prm
 'Then you eat the apple and I promise to eat the pear.'

If (iiia) is followed by (iiib), B's utterance is not acceptable. If (iiia) is followed by (iiic), A's utterance in (iiic) is better. The first conjunct just repeats B's utterance. In this respect, the first conjunct conveys something established in the discourse, not part of the main point of the whole utterance. In some sense, the Mood for the first conjunct is contextually given. There seem to be other constraints in conjoining different types of speech act, but I will leave this for further research.

One interesting thing about *kuliko* is that it does not combine MPs. Instead, a MP can be used with a conjunction connective after another with a full stop in between:

- (10) a. *incho-ka o-ass-ta kuliko mina-ka o-ass-ta.
 Incho-nom come-pst-dec and Mina-nom come-pst-dec
 'Incho came and Mina came.'
 b. incho-ka o-ass-ta. kuliko mina-ka o-ass-ta.
 Incho-nom come-pst-dec and Mina-nom come-pst-dec
 'Incho came. And Mina came.'

The claim that MPs are not conjoined is supported by the fact that *kuliko* 'and' is not used in conjoining embedded MPs:

- (11) *chano-nun incho-ka o-ass-ta kuliko mina-ka o-ass-ta-ko
 Chano-top Incho-nom come-pst-dec and Mina-nom come-pst-dec-cmp
 cwucangha-yess-ta.
 claim-pst-dec
 'Chano claimed that Incho had come and Mina came.'

This also supports the idea that *kuliko* never combines two MPs into one. We need to explain why it does not combine two MPs into one either in embedded clause or in matrix clauses and why it can occur only with one matrix MP.

2.2 Disjunctions of speech acts & MPs

There are also various disjunction connectives:

- (12) a. *-(i)na*: N|NPs (without a case-marker), PPs, AdvPs
 i. incho-na mina ii. kang-eyse-na pata-eyse
 Incho-or Mina river-in-or sea-in
 'Incho or Mina' 'in the river or in the sea'
 iii. honca-na yelesi iv. *mil-ess-ta-na tangki-ess-ta-ko
 alone-or in.crowds push-pst-dec-or pull-pst-dec-cmp
 'alone or in crowds' 'that (he) pushed ... or (he) pulled ...'

- b. *-kena*: Vs|VPs, AspPs, TPs, MPs
- i. [mil-kena tangki]-ess-ta ii. [mil-{ko.iss, ess}-kena tangki-{ko.iss, ess}]-ta
 push-or pull-pst-dec push-{prg, pst}-or pull-prog-pst-dec
 'did push or pull' 'was pushing or pulling'
- iii. mil-ess-ta-kena tangki-ess-ta-ko
 push-pst-dec-or pull-pst-dec-cmp
 'that (he) pushed ... or (he) pulled ...'
- c. *hokun*: NPs (with or without a case-marker), PPs, AdvPs, MPs, CPs
- i. inho-(lul) hokun mina-lul ii. kang-eyse hokun pata-eyse
 Inho-(acc) or Mina-acc river-in or sea-in
 'Inho or Mina' 'in the river or in the sea'
- iii. honca hokun yelesi iv. yenghwa po-ass-nya? hokun sanchayk ka-ss-nya?
 alone or in.crowds movie see-pst-int or walk go-pst-int
 'alone or in crowds' 'Did you see a movie? Or did you go walking?'
- v. yenghwa po-ass-nya-ko hokun sanchayk ka-ss-nya-ko mwul-ess-ta.
 movie see-pst-int-cmp or walk go-pst-int-cmp ask-pst-dec
 'I asked whether you saw a movie or whether you went walking.'

There are correspondence relations between conjunction and disjunction connectives: *-(k)wa* vs. *-na*; *-ko* vs. *-kena*; and *kuliko* vs. *hokun*. Disjunction connectives are less restricted in that *-na* and *-kena* can additionally combine AdvPs and MPs, respectively.

Among the disjunction connectives, only *-kena* and *hokun* can be attached to MPs. A disjunction connective corresponding to *kuliko* is *hokun* 'or' in that both are free morphemes and they can be attached to the same types of phrases. *Hokun* is not used to combine two MPs into one in matrix clauses or in embedded clauses:

- (13) a. *[_{MP} mino-ka sanchaykha-yess-{ta, nya}] hokun [_{MP} pro swuy-ess-{ta, nya?}]
 Mino-nom walk-pst-{dec, int} or rest-pst-{dec, int}
 'Mino walked or he took a rest., Did Mino walk or did he take a rest?'
- b. [_{MP} mino-ka sanchaykha-yess-{*ta, nya?}] hokun [_{MP} swuy-ess-{*ta, nya?}]
 Mino-nom walk-pst-{dec, int} or rest-pst-{dec, int}
 'Mino walked or he took a rest., Did Mino walk or did he take a rest?'
- c. *chano-nun [_{MP}[_{MP} mino-ka sanchaykha-yess-{ta, nya}] hokun
 Chano-top Mino-nom walk-pst-{dec, int} or
 [_{MP} pro swuy-ess-{ta, nya}]]-ko malha-yess-ta.
 rest-pst-{dec, int}-cmp say-pst-dec

'Chano {said that Mino had walked or that he had taken a rest, asked whether Mino had walked or whether he had taken a rest}.

The connective *hokun* is not used to combine two MPs into one, as in (13a). (13b) shows that it can be attached to one matrix MP, but it is allowed for interrogatives, but not for declaratives. (13c) indicates that *hokun* is only attached to a matrix MP.

-kena does not disjoin matrix MPs, as in (14), but it disjoins embedded MPs, as in (15):

- (14) a. *_[MP] Mino-ka sanchaykhay-ss-{ta, nya}]-kena [_[MP] pro swuy-ess-{ta, nya?}].
 Mino-nom walk-pst-{dec, int}-or rest-pst-{dec, int}
 'Mino walked or he took a rest., Did Mino walk or did he take a rest?'
- b. *_[MP] pro sanchaykha-la]-kena [_[MP] pro swuy-la]!
 walk-imp-or rest-imp
 'Walk or take a rest!'
- (15) a. Yena-nun [_[MP] [_[MP] Mino-ka sanchaykhay-ss-{ta, nya}]-kena
 Yena-top Mino-nom walk-pst-{dec, int}-or
 [_[MP] pro swuy-ess-{ta, nya}]]-ko {malha, mwut}-(nu)n-ta.
 rest-pst-{dec, int}-cmp {say, ask}-impf-dec
 'Yena {says, asks} {that, if} Mino walked or {that, if} he took a rest.'
- b. Yena-nun Mino-eykey [_[MP] [_[MP] pro sanchaykha-la]-kena [_[MP] pro swuy-la]]-ko
 Yena-top Mino-to walk-imp-or rest-imp-cmp
 chwungkoha-n-ta.
 advise-impf-dec
 'Yena advises Mino to walk or take a rest.'

This shows that *-kena* is really a disjunction operator that combines two MPs into one. Then the question is why matrix MPs are not disjoined into a larger MPs.

To summarize, two connectives only can be attached to, but not combine, MPs, and only one connective can combine MPs:

- (16) a. *Kuliko* and *hokun* are conjunction and disjunction connective, respectively, that are attached to a matrix MP, but they do not combine MPs into one.
 b. *-kena* is the only disjunction connective that combines embedded MPs into one.
 c. There is no conjunction connective that combines MPs into one.

In the rest of the paper, I will explain these observations. We need to note that there is no conjunction connective that combines MPs into one. The conjunction connective that corresponds to *-kena* is *-ko*, but the latter is not used to combine MPs.⁵⁾

3. Coordination of speech acts

3.1 Conjunction of speech acts

There are various differences between matrix MPs and embedded MPs. A matrix MP forms a speech act because it is uttered as a separate sentence. An embedded MP does not become a speech act because it is not separately uttered. This is confirmed by the fact that only a matrix MP allows a speech-act adverb:

- (17) a. ecey solcikhi mina-ka o-ass-ta.
 yesterday frankly Mina-nom come-pst-dec
 ‘Frankly, Mina came yesterday.’

5) Since *-kena* does not combine matrix MPs, *-ko* is expected not to combine matrix MPs, and this is really the case, as in (9biii). *-ko* does not combine embedded MPs either:

- I. *yena-nun inho-ka wul-ess-ta-ko mina-ka wus-ess-ta-ko
 Yena-top Inho-nom cry-pst-dec-and Mina-nom laugh-pst-dec-cmp
 sayngkakha-yess-ta.
 think-pst-dec
 ‘Yena thought that Inho cried and Mina laughed.’

This might be accidental because *-ko* as a connective is a homonym of the complementizer *-ko*, which is also used with a MP. But this should not be the reason that it does not conjoin MPs, because in Korean there are many homonyms of grammatical morphemes. The expression *-ess-ess* ‘prf-pst’ indicates that the first one is an aspectual morpheme and the second is a tense morpheme. Two homonyms of *-(k)wa* can be repeated:

- ii. na-nun [inho-wa mina]-wa hamkkey kel-ess-ta.
 I-top Inho-and Mina-with together walk-pst-dec
 ‘I walked with Inho and Mina.’

The first *-wa* is a conjunction coordinator that combines two NPs and the second *-wa* is a postposition that is attached to the conjunction of the two NPs. Therefore we expect there is a different reason that *-ko* does not combine embedded MPs. Considering the fact that disjunction connectives are more lenient in the range of phrases they combine, I suppose that it is related to the semantic difference between conjunction and disjunction coordinations, and I will attempt to give a semantic and pragmatic analysis.

- b. ??in_{ho}-nun [_{MP} ecey solcikli mina-ka o-ass-ta]-ko malha-yess-ta.
 Inho-top yesterday frankly Mina-nom come-pst-dec-cmp say-pst-dec
 'Inho said that frankly, Mina came yesterday.'

A matrix MP allows a speech-act adverb, as in (17a). In (17b), where the embedded MP does not denote a speech act, the expression *solcikli* 'frankly' is not allowed in it.

Another difference between matrix MPs and embedded MPs lies in their forms.⁶⁾ For example, *-(e)/y* can be used as a declarative mood marker only in a matrix clause.⁷⁾ And the declarative *-ta* becomes *-la* after the copula *-i* in an embedded MP:

- (18) a. in_{ho}-ka pap-ul mek-ess-{ta, e}.
 Inho-nom meal-acc eat-pst-dec 'Inho ate a meal.'
 b. mina-nun in_{ho}-ka pap-ul mek-ess-{ta, *e}-ko malha-yess-ta.
 Mina-top Inho-nom meal-acc eat-pst-{dec, dec}-cmp say-pst-dec
 'Mina said that Inho ate a meal.'
 c. mina-nun in_{ho}-ka chencay-i-{*ta, la}-ko sayngkakha-n-ta.
 Mina-top Inho-nom genius-be-{dec, dec}-cmp think-npst-dec
 'Mina thinks Inho to be a genius.'

There are also interrogative mood markers not allowed in embedded clauses:

- (19) mina-nun in_{ho}-ka ttena-ss-{nya, *ni, *na, *ul-kka, *upni-kka}-ko
 Mina-top Inho-nom leave-pst-{int, int, int, mod-int, hon-int}-cmp
 mwul-ess-ta.
 ask-pst-dec
 'Mina asked if Inho {left, left, will/would have left, left}.'

The interrogative mood markers *-ni*, *-na*, *-(u)l-kka*, and *-(u)pni-kka* and an imperative mood marker *-(e)la* and *-e* are used only in matrix MPs:

6) We know in English that for a constituent question, the position of a *wh*-phrase in a matrix clause is different from that of a *wh*-phrase in an embedded clause, as pointed out in Rizzi (1997). There is an inversion of the subject and the tensed auxiliary verb in a matrix interrogative, but not in an embedded interrogative.

7) This morpheme can have various forms. If the preceding verb or VP ends with /e/ or /a/, it is not overtly realized. If the preceding verb ends with *ha* 'do', it is realized as *-y*. And it can be used in interrogatives and imperatives too, depending on the intonations.

- (20) a. pap-ul mek-{??ula, ela, e}!
 meal-acc eat-{imp, imp, imp} 'Eat a meal!'
- b. mina-nun inho-eykey pap-ul mek-{ula, *ela, *e}-ko malha-yess-ta.
 Mina-top Inho-to meal-acc eat-{imp, imp, imp}-cmp say-pst-dec
 'Mina told Inho to eat a meal.'

Thus matrix and embedded MPs are distinguished. Matrix MPs are not embedded clauses with implicit structures, and they need to be discussed separately.⁸⁾

We saw that *kuliko* is attached to a matrix MP, but that two MPs are not conjoined into one, as we saw in (10). We also saw that a matrix MP allows a speech-act adverb. There is other evidence that a matrix MP forms a speech act, and that *kuliko* combines speech acts. First, when *kuliko* connects two sentences, each can have a speech-act adverb:⁹⁾

8) I have discussed cases where *kuliko* is attached to a MP. But it is not only used to combine two speech acts, but it is used to combine other phrases like NPs, PPs, AdvPs and CPs. Even when it is used to combine other phrases on the surface, there is a way that it is used to combine speech acts. If a speech-act adverb is put after it, the second conjunct behaves like a separate speech act:

- i. a. inho, kuliko mina-ka keki-ey ka-ss-ta.
 Inho and Mina-also there-at go-pst-dec
 'Inho, and Mina went there.'
- b. ?*inho, kuliko solcikhi mina-ka keki-ey ka-ss-ta.
 Inho and frankly Mina-nom there-at go-pst-dec
 'Inho, and frankly, Mina went there.'
- c. inho-ka, kuliko solcikhi mina-to, keki-ey ka-ss-ta.
 Inho-nom and frankly Mina-also there-at go-pst-dec
 'Inho, and frankly Mina also, went there.'
- ii. inho-ka (keki-ey ka-ss-ta.) kuliko solcikhi mina-to keki-ey ka-ss-ta.
 Inho-nom there-at go-pst-dec and frankly Mina-also there-at go-pst-dec
 'Inho went there. And frankly Mina also went there.'

If the conjunction connective is followed by a NP with a nominative case without a speech-act adverb, the sentence is fine. If a speech-act adverb is added, the sentence with the nominative case maker on the second conjunct becomes odd. This shows that *kuliko* in (ia) is a conjunction connective combining NPs. On the other hand, if the first NP is marked with the nominative case and the second NP is marked with *-to* 'also', as in (ic), the sentence becomes fine. This implies that *kuliko* in this case is a connective in the sentence-level. The sentence is derived from (ii) by some sort of deletion.

9) On the other hand, other conjunction connectives do not allow a speech-act adverb:

- i. a. *inho-wa solcikhi mina-ka keki-ey ka-ss-ta.
 Inho-and frankly Mina-nom there-at go-pst-dec
 'Inho and frankly Mina went there.'
- b. *inho-ka ttena-ss-ko solcikhi mina-ka an o-ass-ta.
 Inho-nom leave-pst-and frankly Mina-nom not come-pst-dec

- (21) solcikhi mina-ka yeypp-e. kuliko taylyakcekulo khi-ka 180cm-i-a.
 frankly Mina-nom pretty-dec and roughly height-nom 180cm-be-dec
 'Frankly, Mina is pretty. And roughly, her height is 180cm.'

Second, when *kuliko* connects two sentences, each can have a separate vocative:

- (22) inho-ya, cikum ttena-la. kuliko, mina-ya, cino-ka o-nun-ci po-ala.
 Inho-voc now leave-imp and Mina-voc Cino-nom come-adn-nml see-imp
 'Inho, leave now! And Mina, go out and see if Cino comes!'

Third, two MPs uttered by different speakers can be coordinated by *kuliko*. In (23), A's and C's utterances are joined with *kuliko*:

- (23) A: na-nun keki-eyse inho-lul manna-ss-e.
 I-top there-at Inho-acc see-pst-dec 'I saw Inho there.'
 B: kulay?
 'Did you?'
 C: kuliko na-nun keki-eyse mina-lul manna-ss-e.
 and I-top there-at Mina-acc see-pst-dec
 'And I saw Mina there.'
 B: kulemyen twul-i keki-eyse teyithuha-yess-na?
 then two-nom there-at date-pst-int
 'Then were the two on a date there?'

Thus we can conclude that *kuliko* combines speech acts. Then we might expect that *kuliko* could combine two different types of speech acts. But it does not:

- (24). a. ??mina-ka ka-ss-e. kuliko inho-to ka-ss-nya?
 Mina-nom go-pst-dec and Inho-also go-pst-int
 'Mina went. And did Inho go?'

'Inho left and, frankly, Mina did not come.'

In this subsection, we will see that *kuliko* is a connective combining speech acts and that there is no connective combining MPs. I do not know if this is an accidental fact. But given that in Korean there are various conjunction connectives, it is plausible to assume that it is not accidental and it needs to be explained.

- b. ??mina-ka ka-ss-e. kuliko inho-nun o-ala!
 Mina-nom go-pst-dec and Inho-top come-imp
 'Mina left. And Inho come!'

A statement does not combine with a question or a command.¹⁰⁾ Thus two conjoined speech acts must be of the same type:

(25) *Kuliko* combines speech acts of the same type.

Speech acts of the same type are coordinated, just as phrases of the same type are.¹¹⁾

Utterance of a MP becomes a speech act, but the meaning of a speech act does not come from a mood marker. We cannot assume any syntactic or semantic difference between a matrix MP and an embedded MP, because some mood markers are used in both MPs. Moreover, the same form can be used as a different speech act:¹²⁾

10) If *kuliko* is replaced with *kulentey* 'but'/'by the way' or *kulenikka* 'therefore', the examples become fine.

- i. a. mina-ka ka-ss-e. kulentey inho-to ka-ss-nya?
 Mina-nom go-pst-dec by.the.way Inho-also go-pst-int
 'Mina went. And {Inho went., ??did Inho go?}
 b. mina-ka ka-ss-e. kulenikka inho-nun o-ala!
 Mina-nom go-pst-dec therefore Inho-top come-imp
 'Mina left. And Inho come!'

The two connectives are not typical ones because phrases that they can conjoin are extremely limited, and I am not interested in peculiar properties of some connectives.

11) There are exceptional cases where two different types of speech act are conjoined:

- i. a. mina-nun onul ttena-ss-e. kuliko inho-to ttena-ss-ci?
 Mina-top today leave-pst-dec and Inho-also leave-pst-int
 'Mina left today. And did Inho leave too?'
 b. inho-ka ka-ss-e. kuliko nwu-ka o-ass-ci?
 Inho-nom go-pst-dec and who-nom come-pst-int
 'Inho left. And who came?'

In these cases, the second speech acts are not provoked for the first time. In (ia), the interrogative is not really a question that the speaker asks because she does not know if Inho left. She is just asking the addressee to confirm what she knows. In (ib), a statement is combined with a question, but the question is not a normal question. It means 'No one came.' or the speaker already knows who came. Therefore, considering only typical uses of matrix MPs, we can maintain the condition given in (25).

12) In general, a MP can be used as an echo question. Here is an echo question of a declarative sentence:

- i. inho-ka o-ass-ta?

- (26) inho-nun cip-ey {ka-ss-ci., ka-ss-ci?, ka-ci!}
 Inho-top home-to go-pst-CI, go-pst-CI, go-CI
 '[Inho is going home., Is Inho going home?, Inho, go home!]

The morpheme *-ci* can be preceded by a Tense, which allows us to assume *-ci* is a mood marker. The sentence with it can form a different speech act, depending on the intonation.¹³⁾ This shows the speaker can decide what speech act she performs with a matrix MP.

The speaker adds something to a matrix MP to form a speech act. I will represent it as *v*. If ϕ is a matrix MP, the speech act we get by a speaker's uttering ϕ is represented as $v(\phi)$. I do not suppose that *v* is structurally realized with a null performative verb, because a performative verb has the effect of performativity only when it is actually uttered. A null form is not actually uttered. Moreover, if a matrix MP is an embedded structure, it should have the same form as an embedded MP. But this is not the case, as shown in (18-20). Therefore I suppose that *v* is the act of uttering.

A matrix MP is uttered in an utterance context, just as an embedded MP is used in a linguistic context provided by the upper clause. *v* allows a speech-act adverb and two different speech acts may involve different speakers or different addressees, as shown in (21-23). And *kuliko* combines speech acts, but not MPs that do not form a speech act:

- (27) a. $v([_{MP} \phi].)$ kuliko $v'([_{MP} \psi].)$
 b. $*v([_{MP} [_{MP} \phi] kuliko [_{MP} \psi]].)$

Even though *kuliko* combines speech acts, it looks as if it were attached to the second MP.

3.2 Disjunction of speech acts

Krifka (2001) claims that two speech acts are not disjoined because there is no context a speaker can make such an utterance felicitously. Take commands for example:

- (28) Pick up the ball or pick up the racket.
 a. 'Act to make true: You pick up the ball or you pick up the racket.'
 b. 'Pick up the ball, or pick up the racket, I don't know which.'

Inho-nom come-pst-dec 'Inho came?'

This can be considered a different type of speech act.

13) We can make the same observations about the mood marker *-e/ø/y*.

The imperative can be interpreted in two ways. In one reading, two alternative actions are given as options as one command as a whole. This is not really a case of disjunction of speech acts. In the other, two commands are disjoined. Krifka claims that this is a case where the speaker has not made up her mind, and that, citing Merin's (1992) idea of 'weak' readings of permission sentences, the sentence is not a command any more.

Krifka (2001) and Szabolcsi (1997) claim questions are not disjoined, while Belnap and Steel (1976) claim they are. Thus it is a controversial issue whether speech acts are disjoined. In resolving this issue, it will be helpful to look at uses of *hokun* 'or'. We saw that *hokun* behaves like *kuliko* in that it does not combine matrix MPs into one but is attached to one MP. Therefore we could claim that *hokun* combines two speech acts:

- (29) a. $v(\phi.)$ *hokun* $v'(\psi.)$
 b. $v(\phi?)$ *hokun* $v'(\psi?)$
 c. $v(\phi!)$ *hokun* $v'(\psi!)$

In this respect, *hokun* is a good test of whether speech acts are disjoined.

According to the test of *hokun*, some speech acts are disjoined, but others are not. To begin with, it is impossible to disjoin two statements:

- (30) ??*inho-ka ttena-ss-ta. hokun mina-ka ttena-ss-ta.*
 Inho-nom leave-pst-dec or Mina-nom leave-pst-dec
 'Inho left. Or Mina left.'

The reason is that the first speech act implicates that the speaker believes that Inho left. This is incompatible with the implicature from the second sentence. The utterance of *hokun* implicates that the speaker does not believe that Inho left (but that Inho might have left). This leads to contradiction, which makes the discourse odd.¹⁴⁾

However, other speech acts can be disjoined. Consider questions first. Questions may, or may not, be disjoined. In (8), two questions are not disjoined, but some questions are:

14) More precisely, the oddness of the discourse can be shown by the verifications of the statements with respect to the speaker's epistemic alternatives. If a speaker utters a declarative sentence ϕ , then the statement must be true in all of the speaker's epistemic alternatives. If the speaker utters another declarative sentence 'or ψ ' as a separate speech act, then it must also be true in all of the epistemic alternatives. Then the two statements are true for the same set of epistemic alternatives and the speaker should have used a conjunction connective. If the disjunction connective is to be felicitously used, there must be epistemic alternatives in which ϕ does not hold and those in which ψ does not hold. These conditions are not met.

- (31) inho-ka ka-ss-nya? hokun mina-ka ka-ss-nya?
 Inho-nom go-pst-int or Mina-nom go-pst-int
 'Did Inho leave? Or did Mina leave?'

Krifka claims that (31) becomes one question which can be answered by “yes” or “no”, but to this question, we cannot answer with *yes* or *no*. They together seem to form a *wh*-question. Then what is the difference between (8) and (31)?

If each question is assumed to partition the context, following Hamblin (1973), Karttunen (1977), Groenendijk & Stokhof (1984), etc., each of the two questions in (8) leads to each column in (32). Each column is a set of possible answers to a question. The two partition structures are incompatible with each other, and the two structures cannot be merged to one so that the two questions can be re-interpreted as a single question. When we try to merge them, none of the original partitions survives.

(32)

Inho went to Seoul	Mina went to Seoul
Inho went to Beijing	Mina went to Beijing
Inho went to Tokyo	Mina went to Tokyo
...	...

In (31), on the other hand, the subjects are focused and contrasted with each other. Thus the two partition structures for the two questions are compatible with each other and can merge into one partition structure, as in (33). In the resulting partition structure, the partitions for “Inho left” and for “Mina left” are maintained. In this context, it is not appropriate to say “yes” or “no”. The context resembles a context for a *wh*-question, and the answer needs to be something like “Inho left”, “Mina left”, or “Neither left”.

Inho left	Mina left
Inho did not leave	Mina did not leave

(33) ⇒

Inho left
Mina left
Neither (Inho nor Mina) left

If two questions are re-interpreted into one question, we can ask if each of the original questions can be regarded as a speech act. It depends on the definition of speech act. In (31), the first question is not canceled or replaced with the second question, as Krifka claims. Each of the two questions contributes to the formation of a complete question. The first question is not a complete question, and the second one is added to complete the speech act the speaker wants to make. In this respect, we need to distinguish incomplete speech acts and complete speech acts. If two speech acts are disjoined, each of them is an incomplete speech act and they combine to form a complete speech act.¹⁵⁾

Imperatives are also disjoined:

(34) A: icye naka-to.toy-e-yo?

now go.out-may-int-hon

'May I go out now?'

B: ani, emma-lul tow-ayahay. tongsayng-ul po-ala! hokun chengso-lul ha-yela!

no mother-acc help-must brother-acc see-imp or sweeping-acc do-imp

'No, you must help Mom. Take care of your brother! Or clean the rooms!'

In this context, B rejected the permission of A's going out, and the first imperative is interpreted as a command. But when the second imperative is added, the first one is not canceled or replaced with the second one. It is just re-interpreted as an option of action, with the second imperative as another option of action. Still the disjunction of the two imperatives as a whole is considered as a command. Each of the imperatives is an incomplete speech act that contributes to the formation of a complete command.

15) One restriction is that incomplete speech acts are not disjoined over different speakers:

i. A: inho-ka ka-ss-nya?

Inho-nom go-pst-int

'Did Inho leave?'

B: ani.

no

A: ??hokun mina-ka ka-ss-nya?

or Mina-nom go-pst-int

'Or did Mina leave?'

If A's first question is answered by B, it becomes a complete speech act, and the use of *hokun* is not appropriate. A's second utterance can be saved if *hokun* is removed or replaced with something like *kulemyen* 'then'. In this case, the second question is also a complete speech act.

In (34), two commands are disjoined to be interpreted as meaning ‘Take care of your brother or clean the rooms!’. About this result, Krifka (2001) claims that the disjunction operator actually applies to two propositions, not to two commands, and that the disjunction of the two propositions is converted to a command, as in (28a). However, there is a problem with Krifka’s claim. Even if a disjunction of two speech acts can be re-interpreted as one speech act with two propositions, we cannot say that the disjunction operator applies to phrases denoting propositions. Each disjunct definitely has the meaning of a speech act, not of a proposition. A disjunction of two imperative MPs is (structurally) not the same as one imperative MP with a disjunction of two smaller phrases in it.¹⁶ Moreover, if a disjunction of MPs is generally re-interpreted as one MP with the disjunction of propositions, we cannot explain why the two questions in (31) are disjoined but those in (8) are not, or why two statements are not disjoined.

The reason that imperatives are disjoined is that an imperative is interpreted with various meanings: a command, permission, request, warning, a plea, a piece of advice, etc. The actual meaning of an imperative is contextually determined, as observed by Schmerling (1982). It means that its modal base and quantificational force are contextually determined. When two imperatives are combined with *hokun*, they can be interpreted as permissions or options of obligatory actions.¹⁷ Even if an imperative is interpreted as a command at first and it is

16) A Tense Phrase is a smaller phrase than a MP and it denotes a proposition.

17) We can say a similar thing about hortatives. Two hortatives can be disjoined:

i inho-ka ka-n-ta-ko kacengha-ca. hokun mina-ka ka-n-ta-ko
 Inho-nom go-impf-dec-cmp assume-hor or Mina-nom go-impf-dec-cmp
 kacengha-ca.
 assume-hor
 ‘Assume that Inho goes. Or assume that Mina goes.’

This is a perfect discourse. The first suggestion is not canceled or replaced with the second and both of the two suggestions make partial contributions to the formation of a complete speech act of suggestion. The whole discourse means ‘Let’s assume that Inho goes or that Mina goes.’

This is possible because a hortative can be interpreted with various meanings. A hortative can be interpreted as a determination, or a proposal, or encouragement, request, advice, etc.:

- ii. a. yenghwa po-ca. (proposal)
 movie see-hot
 ‘Let’s go to see a movie!’
- b. hapkyekha-ca. (determination)
 pass-hor
 ‘Let’s pass the exam!’
- c. mal-lo ha-ca. (request & promise)
 word-with do-hor
 Let’s resolve things with words.

re-interpreted as an option of action, it does not become infelicitous.

We saw two cases where two speech acts are disjoined. Whether speech acts can be disjoined depends on whether a speech act of the same kind is added and the previous speech act can be re-interpreted as an incomplete speech act so that the disjunction of the two speech acts will form a complete speech act. The re-interpretation is possible if one disjunct can be taken to be an alternative option of the complete speech act:

(35) Speech acts are disjoined if one disjunct is interpretable as an alternative of another.

The possibility depends on the semantics and pragmatics of a sentence. Declaratives do not allow the possibility. Some interrogatives do but others do not. Imperatives always do. These can be summarized as follows:

- (36) a. $*v(\phi.) \vee v(\psi.) \neq v((\phi \vee \psi).)$
 b. $v(\phi?) \vee v(\psi?) = v((\phi \vee \psi?))$ if ϕ and ψ are an alternative of each other.
 c. $v(\phi!) \vee v(\psi!) = v((\phi \vee \psi!))$ if ϕ and ψ are an alternative of each other.

4 Coordinations of embedded MPs

4.1 Properties of embedded clauses

In this section, I will discuss embedded MPs. If an embedded clause includes a mood marker, it always forms a *ko*-phrase or adnominal clause, both of which can be assumed to be a CP, depending on whether it is used with a N or V. We are more interested in MPs as complements of verbs. There are two forms of complement clauses of verbs:

-
- d. hwa-ka na-myen simhohup-ul ha-ca. (advice)
 anger-nom arise-if deep.breath-acc do-hor
 'If you get angry, let's take a deep breath!'

A hortative sentence is uttered as a tentative suggestion when the speaker has no authority to decide what to do. One exception is a case of determination where the speaker herself makes a decision. Even in this case, an utterance of a hortative can be just a tentative suggestion of a future action before she makes a final decision. Therefore an additional suggestion can be made to give more options to the addressee (or, the speaker in a case of determination).

- (37) a. [VP [CP [MP α -{ta, nya, (u)la, ca, ma}]-ko] [V β]]
 dec int imp hor prm-cmp
 b. [VP [NP [CP [TP α]-{n, l }]] [N {kes-ul, ci}]] [V β]]
 adn adn thing-acc CI

The semantic differences are discussed in detail in Yeom (2018b).

When a verb takes a complement clause, it can take one or both of the two forms:

- (38) a. inho-nun mina-ka o-{n kes-ul, ass-ta-ko} malha-yess-ta.
 Inho-top Mina-nom come-{adn thing-acc, pst-dec-cmp} say-pst-dec
 ‘Inho told (me) {the fact, $\emptyset$ } that Mina came.’
 b. inho-nun mina-ka o-{n kes-ul, *ass-ta-ko} moll-ass-ta.
 Inho-top Mina-nom come-{adn thing-acc, pst-dec-cmp} not.know-pst-dec
 ‘Inho did not know {the fact, $\emptyset$ } that Mina came.’
 (39) a. inho-nun mina-ka o-ass-{nun-ci, nya-ko} mwul-ess-ta.
 Inho-top Mina-nom come-pst-{adn-nml int-cmp} ask-pst-dec
 ‘Inho asked whether Mina came.’
 b. inho-nun mina-ka o-ass-{nun-ci, *nya-ko}
 Inho-top Mina-nom come-pst-{adn-nml int-cmp}
 {al, kwungkumha}-(y)ess-ta.
 {know, wonder}-pst-dec
 ‘Inho {knew, wondered} whether Mina came.’

A CP headed by *-ko* can be used with a verb that conveys an utterance, in general. However, this is not always the case. A phrase ending with *-ta-ko* can be followed by verbs like *sayngkakha* ‘think’, *mit* ‘believe’, *al* ‘know’, etc.¹⁸) A phrase ending with *-nya-ko* is hardly used with a non-speech act verb, but it seems to be related to the fact that the questioning act involves the addressee. Without the speaker’s explicit utterance, the addressee will not know what the speaker intends to ask. If no addressee is involved, however, a verb that takes a *ko*-phrase does not have to be a conveying verb:

- (40) a. inho-nun tampay-lul kkunh-ca-ko kyelsimha-yess-ta.
 Inho-top tobacco-acc cut-hor-cmp determine-pst-dec
 ‘Inho determined to stop smoking.’

18) In Korean, *al* ‘know’ is not a factive verb when it takes a *ko*-phrase.

- b. inho-nun wenswu-lul kaph.acwu-ma-ko tacimha-yess-ta.
 Inho-top revenge-acc return-prm-cmp determine-pst-dec
 'Inho determined to revenge.'

A hortative and promissive can express a determination if no addressee is involved and no explicit utterance is made. Therefore an embedded MP can be related to various events or states, not just to uttering events.

A common property of *ko*-phrases is that it introduces an opaque non-factual context:

- (41) ??mina-nun ttena-ss-ta. inho-nun mina-ka ttena-ss-ta-ko
 Mina-top leave-pst-dec Inho-top Mina-nom leave-pst-dec-cmp
 {al, sayngkakha}-ko.iss-ta.
 {know, think}-prg-dec
 'Mina left. Inho {knew, thought} that Mina left.'

If the content of the *ko*-phrase is already known to be a fact, the sentence is odd.

Another property of *ko*-phrases is that it is a predicate, as claimed in Yeom (2018a):

- (42) inho-nun (mina-lul) chencay-i-la-ko sayngkakha-n-ta.
 Inho-top Mina-acc genius-be-dec-cmp think-npst-dec
 'Inho thinks of Mina that she is a genius.'
- (43) a. inho-nun (mina-ka nemeci-n kes-ul) yena-ka nemeci-ess-ta-ko
 Inho-top Mina-nom fall-adn thing-acc Yena-nom fall-pst-dec-cmp
 malha-n-ta.
 say-npst-dec
 'Inho says of Mina's falling that Yena fell down.'
- b. inho-nun (mina-ka wus-nun kes-ul) wu(l)-n-ta-ko sayngkakha-n-ta.
 Inho-top Mina-nom laugh-adn thing-acc cry-npst-dec-cmp think-npst-dec
 'Inho thinks of Mina's laughing that she is crying.'

A *ko*-phrase can be predicated of an individual as in (42), or predicated of an actual fact, as shown in (43a-b). In many cases, the objects of the verbs may not occur for various reasons, and one reason is that the fact is not known yet. An embedded interrogative MP can be analyzed in a similar way, and in such a case the fact is normally not known yet.

One difficulty in dealing with *ko*-phrases is that the predication holds between two different contexts. Consider the following:

(44) John thought Mary to be Jessica.

Mary is taken to be the subject of the predicate *be Jessica*, but the former is interpreted in the actual world and the latter in John's belief worlds. In this case, we can think of an individual in John's belief worlds that corresponds to Mary in the actual world and the predicate *be Jessica* applies to that corresponding individual. In some sense, the predication between the two expressions is indirect. A similar relation is involved in the interpretation of (43a-b). A fact in the actual world expressed by a *kes*-phrase is predicated of by a proposition denoted by a *kes*-phrase. I explain this in detail in the Appendix.

4.2 Disjunction of embedded MPs

We have a disjunction connective *-kena* joining embedded MPs, as shown in (6). One interesting fact about *-kena* 'or' is that when it disjoins MPs, it is interpreted with the meaning of conjunction. Consider the sentences in (6). (6a) implicates that Yena said that Mino walked and she also said that he took a rest. We can say similar things about (6b) and (6c). This does not mean that the disjunction connective becomes a conjunction connective. It is like a free choice effect:

- (45) You may eat an apple or a pear.
 → You may eat an apple & You may eat a pear.
 -/> You may eat both an apple and a pear

(45) implicates that the addressee is allowed to eat an apple and that she is allowed to eat a pear. But this does not mean that the addressee is allowed to eat both. These effects arise when there is a quantifier over a disjunction connective:

- (46) $Q(\lambda x[A(x) \vee B(x)])$ implicates $\exists xA(x)$ and $\exists xB(x)$.

In (45), *may* is an existential quantifier over deontic alternatives. A quantifier introduces multiple members in the quantification domain, and some members hold of one disjunct and others hold of the other disjunct. These are called existence implicatures in Yeom (2017).

As existence implicatures, (47) implicates that there was an event of Mino's asking my name and another event of his asking my age:

- (47) Mino-nun kunye-uy ilum-ina nai-lul mwul-ess-ta.
 Mino-top she-gen name-or age-acc ask-pst-dec
 'Mino asked my name or age.'
 → Mino asked my name & he asked my age.
 -/> There was one event of Mino's asking both my name and age.

Therefore, there was more than one event of asking. (47) does not mean that at one particular time there is one event of asking either the name or the age.

This also applies to (15a), and we need to assume that there is an implicit quantifier over the disjunction connective. Otherwise, the sentence becomes odd:¹⁹⁾

- (48) ??kuttay yena-nun [_{MP}[_{MP} mino-ka ttwuy-ess-ta]-kena
 then Yena-top Mino-nom run-pst-dec-or
 [_{MP} pro kel-ess-ta]]-ko malha-yess-ta.
 walk-pst-dec-cmp say-pst-dec
 'Then Yena said that Mino had run or that he had walked.'
 (49) ??kuttay inho-nun mina-ka pappu-ta-kena yena-ka phikonha-ta-ko
 then Inho-top Mina-nom be.busy-dec-or Yena-nom be.tired-dec-cmp
 sayngkakha-yess-ta.
 think-prg-dec
 'At that time Inho thought that Mina was busy or Yena was tired.'

With the addition of *kuttay* 'then', we can assume that there was only one event of saying and only one state of thinking, respectively, and under those assumptions, the sentences become odd. If the sentences are to have the meaning of opting for either of the two actions or states at the time of saying or thinking, there needs to be a quantifier (or, modal operator) like *-(u)l.kes.i* between the time expression and the disjunction structure:

19) If the sentence is negated, the sentence becomes fine:

- i. ??inho-nun mina-ka pappu-ta-kena yena-ka phikonha-ta-ko sayngkakha-ci
 Inho-top Mina-nom be.busy-dec-or Yena-nom be.tired-dec-cmp think-nml
 anh-ass-ta.
 not.do-pst-dec
 'Inho did not think that Mina was busy or Yena was tired.'

This is because Inho did not believe either of the two propositions. There is no need of any event or state of believing.

- (50) kuttay yena-nun [_{MP}[_{MP} mino-ka sanchaykha-yess-ta]-kena [_{MP} pro
 then Yena-top Mino-nom walk-pst-dec-or
 swuy-ess-ta]]-ko malha-l.kes.i-ta.
 rest-pst-dec-cmp say-mod-dec
 'Then Yena will say that Mino walked or he took a rest.'
- (51) kuttay inho-nun mina-ka pappu-ta-kena yena-ka phikonha-ta-ko
 then Inho-top Mina-nom be.busy-dec-or Yena-nom be.tired-dec-cmp
 sayngkakha-yess-ul.kes.i-ta.
 think-pst-mod-dec
 'At that time Inho would think that Mina was busy or Yena was tired.'

Here the modal operator *ul.kes.i* introduce possible worlds in the quantification domain and in some of them there is an event of saying or a state of thinking involving one MP and in others there is also an event of saying or a state of thinking involving the other MP. Still it can be maintained that each MP is interpreted with respect to one event of saying or a state of thinking in each possible world. That is, (51) implicates that Inho may think that Mina was busy or he may think that Yena was tired.

These observations are rather surprising because the verb *sayngkakha* 'think' is also a quantifier over Inho's doxastic alternatives. Each of the two MPs could hold in some doxastic alternatives or others. Such interpretations really arise when smaller phrases than MPs are disjoined:

- (52) inho-nun mina-na yena-ka aphu-ta-ko sayngkakha-ko.iss-ta.
 Inho-top Mina-or Yena-nom be.sick-dec-cmp think-prg-dec
 'Inho thinks that Mina or Yena is sick.'

By this statement, it is implicated that Inho thinks that Mina might be sick and that Yena might be sick. However, this kind of inference does not arise from (49) (or (51)).

This surprising fact can be explained by claiming that a MP is related to a certain event or state involving the meaning of the MP. As for an embedded MP, generally the matrix verb introduces an event or state related to the meaning of the MP. However, the MP might not be related to the matrix verb, as discussed in Yeom (2018a):

- (53) yena-nun mina-ka hayngpokha-ta-ko malha-yess-ta.
 Yena-top Mina-nom be.happy-dec-cmp say-pst-dec
 'Yena said that Mina was happy.'

- (54) yena-nun mina-ka hayngpokha-ta-ko sayngkakha-yess-ta.
 Yena-top Mina-nom be.happy-dec-cmp think-pst-dec
 'Yena said that Mina was happy.'
- (55) inho-nun [_{MP} mina-ka hayngpokha-ta]-nun kes-ul mit-(ci anh)-ass-ta.
 Inho-top Mina-nom be.happy-dec-adn thing-acc believe-nml not.do-pst-dec
 'Inho did not believe that Mina

(55) can be preceded by (53), but not by (54). In (55), the situation relevant to the embedded MP is not the situation of believing or not believing in the matrix clause but the situation of saying in (53). In (55), the MP is embedded in an adnominal clause and the *kes*-phrase denotes a proposition. The *kes*-phrase is licensed by a preceding event of someone else's assertion. That is why (53) can precede (55). (54) also introduces a state that is relevant to a MP, but it is not relevant to the MP in (55) because the existence of a proposition is not known to Inho when the proposition is simply what someone else thinks.

Without the declarative mood marker *-ta*, a *kes*-phrase can denote a fact and there is no need of any preceding event related to a proposition:

- (56) inho-nun mina-ka pwulhayngha-n kes-ul al-ass-ta.
 Inho-top Mina-nom be.unhappy-adn thing-acc know-pst-dec
 'Inho knew that Mina was unhappy.'

This can be preceded by either (53) or (54), because a fact-denoting *kes*-phrase itself does not need any event that involves the fact. This supports the idea that a MP is closely related to an event or state denoted by a predicate. We can suppose that a use of a MP is licensed by an event or state that involves the proposition denoted by that MP.

There is another piece of evidence that a MP is interpreted in relation to an event involving a proposition. According to Ross (1970), when a matrix MP is uttered, the utterance context provides linguistic antecedents for reflexives:

- (57) a. As for myself, I won't be invited.
 b. *As for himself, he won't be invited.

The speaker herself licenses the use of the reflexive. This phenomenon also arises in embedded clauses, with an event of saying introduced by the predicate in the matrix clause. The agent of the event licenses a use of a reflexive in the embedded clause:

- (58) a. Jules said that as for himself he wouldn't be invited.
 b. I said that as for myself I wouldn't be invited.

A use of a reflexive does not just depend on whose perspective a clause describes. The antecedent of a reflexive needs to be a participant of an event provided in the context:

- (59) a. According to Mary, John is a little taller than herself.
 b. *As for Mary, John is a little taller than herself.

In (59a), Mary is not specified, but implied, as the speaker of the whole sentence. Thus we can think of an utterance situation, which allows the use of the reflexive herself. In (59b), however, we could suppose that the sentence is described from the perspective of Mary, but this is not allowed if no event is implicated.²⁰⁾

20) Lasersohn (2005) and Stephenson (2007) claim the index of judge is necessary to explain the restrictions that some predicates show. In Korean, the verb *coh* has two uses: as a one-place predicate and as a two-place predicate. When it is used as a two-place predicate in a matrix clause, the subject needs to be the speaker:

- i. a. na-nun mina-ka coh-ta.
 I-top Mina-nom like-dec 'I like Mina.'
 b. ??cina-nun mina-ka coh-ta.
 Cina-top Mina-nom like-dec 'Inho told Mina that Cina liked her.'

The same restriction is also observed in embedded clauses:

- ii. a. inho-ka mina-eykey caki-nun mina-ka coh-ta-ko malha-yess-ta.
 Inho-nom Mina-to self-top Mina-nom like-dec-cmp say-pst-dec
 'Inho told Mina that he liked her.'
 b. ??inho-ka mina-eykey cina-nun mina-ka coh-ta-ko malha-yess-ta.
 Inho-nom Mina-to Cina-top Mina-nom like-dec-cmp say-pst-dec
 'Inho told Mina that Cina liked her.'

The matrix verb is a conveying verb, and the use of *coh* is fine when the subject of the embedded clause is the speaker of the conveying verb. This does not seem to be a pragmatic restriction. Even if there is an utterance implicated, the condition is not met:

- iii. a. ??inho-nun mina-eykeyse caki-nun inho-ka coh-ta-ko tul-ess-ta.
 Inho-nom Mina-to self-top Inho-nom like-dec-cmp hear-pst-dec
 'Inho heard from Mina that she liked him.'
 b. ??inho-nun mina-eykeyse cina-nun mina-ka coh-ta-ko tul-ess-ta.
 Inho-nom Mina-to Cina-top Mina-nom like-dec-cmp hear-pst-dec
 'Inho heard from Mina that Cina liked him.'

From the discussions so far, we can claim that a use of a MP needs an event or state that involves the meaning of that MP. For simplicity's sake I will call an event or state a situation.²¹⁾ The condition can be specified as follows:

(60) A use of a MP presupposes a situation that involves the meaning of the MP.

The situation does not need to be structurally realized. The only requirement is that a situation be given in the context for the use of a MP.²²⁾ For a matrix MP, the situation relevant is the actual utterance event, which is not realized structurally, and an embedded MP is generally related to a situation of any type provided by an upper clause. Therefore, the meaning of a MP needs to be defined in such a way:

- (61) a. $\llbracket \alpha\text{-ta} \rrbracket^{c,w} = \exists p[\llbracket \langle P(e, a_1, \dots, a_n, p) \rangle \rrbracket \wedge p = \lambda w' \llbracket \alpha \rrbracket^{c,w'} \wedge p(w)]$
 - $\llbracket \langle \rangle \rrbracket$: presuppositions from the use of a MP.
 - The situation e determines $a_1, \dots, a_n$, and p .
 - If e is an utterance event, e is part of w . ($e \sqsubset w$)
- b. $\llbracket \alpha\text{-nya} \rrbracket^{c,w} = \exists Q[\llbracket \langle P(e, a_1, \dots, a_n, Q) \rangle \rrbracket \wedge Q = \{ \lambda w' \llbracket \alpha \rrbracket^{c,w'}, \lambda w' [\neg \llbracket \alpha \rrbracket^{c,w'}] \} \wedge Q(w)]$ (Yes-No Q)
 - $Q(w)$ means 'which proposition in Q is true in w '
- c. $\llbracket \alpha\text{-nya}_i \rrbracket^{c,w,g} = \exists Q[\llbracket \langle P(e, a_1, \dots, a_n, Q) \rangle \rrbracket \wedge Q = \{ \lambda y \lambda w' [\llbracket \alpha \rrbracket^{c,g[y/i],w'}](x) \mid x \in D_{\tau(x)} \} \wedge Q(w)]$ (Wh-Q)
 - $D_{\tau(x)}$ is the discourse domain of the type of x .

In (61), the variables $e, a_1, \dots, a_n$ are free, and they are assumed to be given by the context. The meaning of a declarative is a proposition and $p(w)$ means that p is true in w . The variable p is bound by the existential quantifier, but its value is actually fully specified in the meaning representation.²³⁾ The interrogative in (61b) is a *Yes-No* question, and the one in (61c) is a

The use of the verb *tut/tul* 'hear' presupposes that there was an utterance but this does not make the utterance acceptable. This shows that *coh* as a two-place predicate can be used only when the experiencer of the verb is the agent of a conveying verb in the upper clause. This indicates that an embedded clause is interpreted with respect to a local context in which it needs to be specified who is the conveyer of the statement. This also shows that a clause needs to be interpreted with respect to an event or state.

21) When I talk about an event or state, it is introduced by a verb. But a better notion is a situation described by a clause, not an event or state described by a VP.

22) In this respect, I do not assume a syntactic structure for speech acts by Speas and Tenny (2003). On the other hand, this does not argue for or against Evaluation Phrases or Evidential Phrases by Cinque (1999).

constituent question. The meaning of a question is a set of propositions and $Q(w)$ is a set of true propositions/answers in w . The parts in $\ll \gg$ are presuppositions. If the MP is a matrix one, the presupposition is satisfied by an utterance event, but if the MP is an embedded one, the presupposition is satisfied by a situation introduced in the upper clause or in a previous context. One difference between a matrix and embedded MP is that a matrix MP is interpreted with respect to the actual world with the presupposition of an utterance event, and that event is part of the actual world. If an embedded MP is interpreted with respect to a possible world accessible from the actual world with the presupposition of a situation, the presupposition can be satisfied in the actual world and that situation is not part of the possible world accessible from the actual world.

In order for a matrix MP to be used felicitously, $\mathcal{U}(\phi)$ needs to satisfy the presuppositions for ϕ . Therefore a speech act needs to be at least interpreted as follows:

$$(62) \ll \mathcal{U}(\phi) \gg^{c,w^*} = \exists \pi [\text{SPEECH-ACT}(u,s,a,\pi) \wedge \pi = \ll \phi \gg^{c,u} \wedge \pi(w^*)]$$

u: utterance event; w^* : the actual world; $u \sqsubset w^*$

“SPEECH-ACT” can be TELL or ASK, depending on the sentence type of ϕ and the speaker’s intention. In the meaning of a speech act, the utterance event is part of the actual world and the MP is interpreted with respect to the actual world. The event also involves the speaker and the addressee. The meaning of a matrix MP as an argument of the speech act predicate is an intensional semantic entity, but it is interpreted with respect to the actual world w^* . The extensional meaning is represented as $\pi(w^*)$. The presuppositions from the use of ϕ can be satisfied by the meanings in (62).

Under the assumption that a MP is interpreted with the presupposition of a situation, a disjunction of two MPs can be interpreted as follows:

$$(63) \ll [\text{MP}[\text{MP } \varphi] \vee [\text{MP } \psi]] \gg^{c,w} \\ = \ll [\text{MP } \varphi] \gg^{c,w} \vee \ll [\text{MP } \psi] \gg^{c,w}$$

The disjunction of two MPs presupposes a situation for the use of it. But each of the smaller MPs also presupposes a situation for the use of them. The context only provides one situation

23) The meaning representation might be confusing because the variable p is existentially bound, but the value is given in the end. The purpose is to separate the presupposition and the assertion. Without the distinction, the representation could be like this:

i. $P(e,a_1,\dots,a_n, \lambda w' \ll a \gg^{c,w'}) \wedge \ll a \gg^{c,w} = 1$

for the use of the disjunction of MPs so that the two smaller MPs are related to that same situation. Then there is indeterminacy between the two smaller MPs at one situation. This is unacceptable. An utterance of a matrix MP is an action, which is based on the speaker's decision to do it. If the speaker is in indeterminacy, she is not in a state to be able to do an action. If she does, she is behaving in a contradictory way. That is, she is not in a position to perform a speech act and every discourse participant knows it from the use of a disjunction structure, but she is behaving as if she were. We can say the same thing about disjunction of embedded MPs. This explains why (48) and (49) are bad.²⁴

However, a disjunction of embedded MPs is acceptable when the disjunction operator has wide scope over the predicate that provides a situation involving the meaning of a MP. According to Larson (1985), Schwarz (1999), etc., a disjunction operator can have wider scope than the one in the surface form. In (50) and (51), the disjunction operator has wide scope over the verb that introduces an event/state, and more than one event/state is involved. (50) is understood as meaning 'Then Yena will either say that Mino walked or say that he took a rest.' There is a different event of saying for each MP in each possible world in the quantification domain for the modal operator *will*.

5 Other effects from assuming a situation

5.1 Conjunctions of MPs

I showed that the use of a MP presupposes the existence of a single situation that involves it. This idea also allows us to explain why matrix MPs are not conjoined. For a felicitous use of a matrix MP, there is one event of utterance. If two MPs are conjoined, the conjunction of the MPs is interpreted as follows:

$$\begin{aligned} (64) \quad & \llbracket [\text{MP} [\text{MP } \varphi] \wedge [\text{MP } \psi]] \rrbracket^{c,w} \\ &= \llbracket [\text{MP } \varphi] \rrbracket^{c,w} \wedge \llbracket [\text{MP } \psi] \rrbracket^{c,w} \end{aligned}$$

If MPs are conjoined, the whole MP is interpreted with the presupposition of a single situation, and the two smaller MPs in it must also be interpreted with respect to the same situation. The two MPs must be uttered simultaneously, which is impossible. You might claim

24) One acceptable interpretation is that the speaker does not know what Inho thought but she knows that Inho had a certain thought. In this case, the disjunction operator has widest scope. This reading is not allowed without an epistemic modality operator.

that each of the MPs is uttered at a smaller situation and that the sum of the two situations forms a larger situation. But the sum of two situations is not a single event of utterance, which violates the condition that each MP is related to a single situation.

We might think of a possibility that a conjunction connective has wide scope over a predicate that provides a situation. That is, (65a) might be interpreted as meaning (65b):

- (65) a. *yena-nun inho-ka wul-ess-ta-ko mina-ka wus-ess-ta-ko
 Yena-top Inho-nom cry-pst-dec-cmp Mina-nom laugh-pst-dec-cmp
 sayngkakha-yess-ta.
 think-pst-dec
 'Yena thought that Inho cried and Mina laughed.'
- b. yena-nun inho-ka wul-ess-ta-ko sayngkakha-ko mina-ka
 Yena-top Inho-nom cry-pst-dec-cmp think-and Mina-nom
 wus-ess-ta-ko sayngkakha-yess-ta.
 laugh-pst-dec-cmp think-pst-dec
 'Yena thought that Inho cried and thought that Mina laughed.'

In this reading, there is more than one situation involved, and in each situation there is one MP involved. However, this reading would be expressed by a disjunction of two MPs, because the two MPs are not simultaneously used in a single situation:

- (66) yena-nun inho-ka wul-ess-ta-kena mina-ka wus-ess-ta-ko
 Yena-top Inho-nom cry-pst-dec-or Mina-nom laugh-pst-dec
 sayngkakha-yess-ta.
 think-pst-dec
 'Yena thought that Inho cried or that Mina laughed.'
 (= 'Yena thought that Inho cried and that Mina laughed.')

This implies that it has no motivation to use a conjunction operator to get this meaning.

This can be compared with a case where a conjunction connective actually combines phrases larger than MPs:

- (67) inho-nun mina-ka pappu-ta-ko, kuliko solcikli yena-ka
 Inho-top Mina-nom be.busy-dec-cmp and frankly Yena-nom
 phikonha-ta-ko sayngkakha-yess-ta.
 be.tired-dec-cmp think-pst-dec

‘Inho thought that Mina was busy, and, frankly, that Yena was tired.’

Even though we have one matrix verb, we can assume that there were two situations of thinking, one related to the first CP and the other to the second. This is clear from the use of the speech-act adverb. We can assume that this is derived from the following:

- (68) inho-nun mina-ka pappu-ta-ko sayngkakha-yess-ta. kuliko solcikli
 Inho-top Mina-nom be.busy-dec-cmp think-pst-dec and frankly
 inho-nun yena-ka phikonha-ta-ko sayngkakha-yess-ta.
 Inho-top Yena-nom be.tired-dec-cmp think-pst-dec
 ‘Inho thought that Mina was busy and, frankly Inho thought that Yena was tired.’

In (68), the two MPs are not interpreted with the presupposition of one and the same situation because the two MPs do not constitute one MP:

- (69) $\llbracket [\text{CP} [\text{CP } \varphi\text{-ko}] \wedge [\text{CP } \psi\text{-ko}]] \rrbracket^{c,w}$
 $= \llbracket \text{ko} \rrbracket (\llbracket [\text{MP } \varphi] \rrbracket^{c,w}) \wedge \llbracket \text{ko} \rrbracket (\llbracket [\text{MP } \psi] \rrbracket^{c,w})$ (a situation for ϕ and another for ψ)
 (Assumption: The meaning of *-ko* takes the meaning of a MP.)

If two *ko*-phrases are conjoined, the two MPs do not have to be interpreted with respect to one situation. This makes (67) acceptable.

5.2 Speech act adverbs

If we introduce the presupposition of an utterance event in interpreting a matrix MP, we can treat some uses of speech-act adverbs easily. Schreiber (1972) and Piñón (2013) considered a speech-act adverb a manner adverb. This idea can apply to (17a):²⁵

- (70) $\llbracket \text{ecey solcikli mina-ka o-ass-ta} \rrbracket^{c,w^*}$
 $= \exists p[\llbracket \text{TELL}(e,s,a,p) \rrbracket \wedge \exists x[\text{manner}^{w^*}(x,e) \wedge \text{frank}^{w^*}(x)] \wedge$
 $p = \lambda w[\text{pretty}^w(\text{mina})] \wedge p(w^*)]$

25) The meaning of the speech act adverb is not part of the truth conditions of the sentence, but I will ignore this in the meaning representation.

However, such an analysis is not general enough to apply to all speech-act adverbials, as pointed out by Mittwoch (1977):

- (71) a. The Smiths are back, because I saw the milkman call at their house this morning.
 = The Smiths are back, (I can tell you this) because I saw the milkman call at their house this morning.
 b. The Smiths are away, in case you are wondering why you haven't heard from them.
 = The Smiths are away, (I tell you this) in case you are wondering why you haven't heard from them.
 c. Are the Smiths hack, since you are their neighbour?
 = Are the Smiths hack, (I ask you this) since you are their neighbour?

These are examples of adverbial clauses modifying speech acts, but they are not manner adverbials for the utterance events. Still speech act events, which are expressed in parentheses, are in the relations expressed by the prepositions *because*, *in case*, and *since* to the propositions denoted by the embedded clauses.²⁶⁾

On the other hand, simply presupposing an event involving the meaning of a MP cannot capture the fact that only matrix MPs can occur with speech-act adverbs, as shown in (17). The embedded MP in (17b) might be interpreted as in (72):

- (17) b. ??incho-nun ecey solcikli mina-ka o-ass-ta-ko malha-yess-ta.
 Incho-top yesterday frankly Mina-nom come-pst-dec-cmp say-pst-dec
 'Incho said that frankly, Mina came yesterday.'
 (72) $\exists p[\ll \exists e[\text{SAID}^*(e, \text{incho}, s, p)] \gg \wedge p = \lambda w' \ll ecey \text{ solcikli mina-ka o-ass-ta } \gg^{c, w'} \wedge \forall w'' \in \text{ACC}(w'')[p(w'')]]$

In (17b), the embedded MP is interpreted, presupposing that there is an event of saying that Mina came yesterday. Still the use of the speech act adverb is not acceptable in the embedded clause. We need to explain why the event of saying does not licence the adverb.

One special thing about a matrix MP is that it is interpreted with respect to the actual world and the situation presupposed from the use of that MP is also part of the actual world, as I mentioned in relation to (61). Therefore when the MP is interpreted, the utterance event is always available. This makes the use of a speech-act adverb acceptable. On the other hand,

26) It is also possible to assume a different predicate than the one representing the speech act. In interpreting embedded MPs, we should consider various predicates. But I will not go into this issue. It needs more extensive discussions to see how speech-act adverbials are interpreted.

if an embedded MP is interpreted with respect to a possible world accessible from the actual world, an event is presupposed that belongs to the actual world. The event is not part of the accessible world. Therefore, it cannot be exploited in interpreting an adverb in the embedded MP. This makes a speech-act adverb unacceptable in an embedded clause.

6. Conclusion

In Korean, MPs are not conjoined to form a larger MP in matrix or embedded clauses. The reason is that each MP is interpreted with the presupposition of a single event or state introduced by an upper clause or the context. If two MPs are conjoined to form a MP, the three MPs should be related to one and the same event or state, but two smaller MPs cannot be interpreted with respect to the single situation simultaneously.

On the other hand, *kuliko* can be attached only to one matrix MP on the surface. This is possible because it is a conjunction operator for speech acts. Since each speech act is an utterance of one matrix MP, each speech act conjoined by *kuliko* includes a separate MP with a separate utterance event.

If two MPs are disjoined to form a larger MP in matrix or embedded clauses, the two MPs need to be related to a single event or state, which is a situation of indeterminacy. In such a context, the speaker may not make an utterance. To avoid such indeterminacy, the disjunction operator may have wide scope over a predicate that introduces a situation related to each MP. This is possible only when the MPs are embedded, and in such a case *-kena* 'or' is used. Interestingly, the disjunction operator implicates the meaning of conjunction. I suppose that this obviates the need to use a conjunction operator that combines two MPs and gets wide scope over a predicate that introduces a situation.

Hokun 'or' can be attached to an imperative or interrogative, but not to a declarative. It does not combine two MPs to a larger MP because it only combines speech acts. And when it combines two speech acts, the utterance of each MP forms an incomplete speech act, and the disjunction of two speech acts forms a complete speech act.

<References>

- Belnap, Nuel D., and Thomas B. Steel. 1976. *The Logic of Questions and Answers*. Yale University Press, New Haven.
- Cinque, Guglielmo. 1999. *Adverbs and Functional Heads*. Oxford: Oxford University Press.
- Emonds, J. E. 1969. *Roots and structure-preserving transformations*. Doctoral dissertation, MIT, Cambridge, MA.
- Groenendijk, J. and Stokhof, M. 1984. *Studies on the Semantics of Questions and the Pragmatics of Answers*. Ph.D. thesis, University of Amsterdam.
- Hamblin, C. 1973. Questions in Montague English. *Foundations of Language* 10(1), 41-53.
- Heycock, C. 2006. Embedded root phenomena. In E. Martin & H. van Riemsdijk (Eds.), *The Blackwell Companion to Syntax* (Vol. II, pp. 174-209). Oxford: Blackwell.
- Hooper, J., & Thompson, S. 1973. On the applicability of root transformations. *Linguistic Inquiry*, 465-497.
- Karttunen, L. 1977. Syntax and semantics of questions. *Linguistics and Philosophy* 1(1), 3-44.
- Kim, Yenghi. 1988. Tungwui cepsokmwunuy thongsa thukseng (Syntactic properties of coordinated sentences in Korean). *Hangeul (Korean Alphabet)* 201/202, 83-118.
- Koh, Jae-sol. 2003. On the Well-formedness Conditions and the Structure of Coordinate Sentences in Korean. *The Linguistic Association of Korea Journal*, 11(2), 135-156.
- Krifka, Manfred. 2001. Quantifying into question acts. *Natural Language Semantics*, 9, 1-40.
- Kwon, Jayil. 1985. *Kwukeuy pokhapmwun kwuseng yenkwu (Study on structures of complex sentences in Korean)*. Seoul: Cipmwuntang.
- Larson, R. 1985. On the Syntax of Disjunction Scope. *Natural Language and Linguistic Theory* 3, 217-264.
- Laserson, Peter. 2006. Relative Truth, Speaker Commitment, and Control of Implicit Arguments. Paper presented at the 37th Meeting of the North East Linguistic Society, University of Illinois, Urbana-Champaign.
- Lee, Iksep, and Hongpin Im. 1983. *Kwukemwunpeplon (Syntax of Korean)*. Seoul: Hakyensa.
- Merin, Arthur. 1992. Permission Sentences Stand in the Way of Boolean and Other Lattice-Theoretic Semantics. *Journal of Semantics* 9, 95-162.
- Mittwoch, Anita. 1977. How to Refer to One's Own Words: Speech-Act Modifying Adverbials and the Performative Analysis. *Journal of Linguistics*, 13.2, 177-189.
- Piñón, Christopher. 2013. Speech-act adverbs as manner adverbs. Université Lille 3 / UMR 8163 STL.
- Rizzi, Luigi. 1997. The Fine structure of the left periphery. In *Elements of Grammar*. Liliane

- Haegeman (ed.): 281-337. Dordrecht: Kluwer.
- Ross, John R. 1970. On declarative sentences. In: Jacobs R.A. & P.S. Rosenbaum, (eds), *Readings in English Transformational Grammar*. Waltham, Mass: Ginn & Co, 222-272.
- Schmerling, Susan. 1982. How imperatives are special and how they aren't. *Papers from the Parasession on Nondeclaratives: Chicago Linguistic Society*, ed. by Robinson Schneider, Kevin Tuite, and Robert Chametzky, 202-218.
- Schreiber, P. A. 1972. Style disjuncts and the performative analysis. *Linguistic Inquiry* 3, 321-347.
- Schwarz, B. 1999. On the Syntax of Either...Or. *Natural Language and Linguistic Theory* 17, 339-370.
- Speas, Margaret, and Carol Tenny. 2003. Configurational Properties of Point of View Roles. pp.p. 315-344 in *Asymmetry in Grammar*, edited by Anna-Maria Di Sciullo. Amsterdam: John Benjamins.
- Szabolcsi, Anna. 1993. Quantifiers in Pair-List Readings and the Non-uniformity of Quantification. *Ninth Amsterdam Colloquium*, pp. 645-664. ILLC/University of Amsterdam.
- Stephenson, Tamina C. 2007. *Towards a Theory of Subjective Meaning*. doctoral dissertation. MIT.
- Yeom, Jae-Il. 2017. Existence implicatures. ms. (available from ResearchGate).
- Yeom, Jae-Il. 2018a. Embedded declaratives in Korean. *Language and Information*, 22-1, 1-27.
- Yeom, Jae-Il. 2018b. Embedded imperatives and command performatives with an embedded clause in Korean. *Language and Information*, 22-2, 27-54.
- Ywu, Hyenkyeng. 1986. Kwuke cepsokmwunuy thongsacek thukciley tayhaye (On syntactic properties of sentences with coordinators or subordinators in Korean). *Hangeul (Korean Alphabet)* 191, 77-104.

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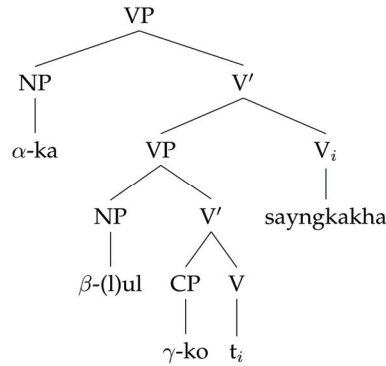
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Appendix

To interpret a *ko*-phrase, I assume a verb at the DS position takes the *ko*-phrase and then moves to a higher V position to take the individual argument:

I.



Now I will explain how this structure is interpreted. To deal with two different contexts, I assume that a variable ranges over intensional semantic entities. To deal with the cases in (42-43), I assume that the verb *sayngkakha* ‘think’ has two meaning specifications, as given in (IIa), and a *ko*-phrase also has two interpretations given in (IIb), depending on whether the *ko*-phrase is predicated of an individual or a fact. In the actual meaning specifications, individual concepts and propositions are used instead of individuals and facts because we need meanings that are defined over two different contexts. Moreover, it is easier to capture meaning relations by propositions rather than by facts. To make this work, we need to assume that a possible world is a set of propositions:

- II. a. $\llbracket \text{sayngkakha} \rrbracket^w = i) \lambda P \lambda y \lambda x \lambda e \exists z [z = y^w \wedge \text{think}(e, x^w, \lambda w' [P^{w'}(y^w)])]$
 - y ranges over individual concepts, z ranges over objects.
 ii) $\lambda P \lambda p \lambda x \lambda e \exists p' [p(w) \wedge p \subset p' \wedge \text{think}(e, x^w, \lambda w' [P^{w'}(p')])]$
- b. i) $\llbracket \gamma\text{-ko} \rrbracket^{c,w,g} = \lambda w' \lambda z [\llbracket \gamma \rrbracket^{c,w',g}(z)]$
 ii) $\llbracket \gamma\text{-ko} \rrbracket^{c,w,g} = \lambda w' \lambda p' [\llbracket \gamma \rrbracket^{c,g} \subset p' \wedge \llbracket \gamma \rrbracket^{c,g}(w')]$
 - utterance context $c = \langle s, a, t^* \rangle$ (s : speaker, a : addressee, t^* : utterance time)

In (IIai) the verb *sangkakha* ‘think’ takes a predicate, which is a *ko*-phrase, and then an individual concept y . For an individual concept, an individual in one possible world is an epistemic alternative corresponding to an individual in another possible world. Then the predicate is predicated of the individual in a possible worlds in the belief context. In (IIaii), the predication relation is more indirect, because we cannot think of a direct correspondence relation between propositions in two contexts. Instead, two propositions are mediated by a weaker proposition p' which is entailed by the two propositions. The verb takes a predicate of propositions and a proposition. In the belief context, the predicate is predicated of a weaker proposition than the one the verb actually takes. For each of these two cases, the meaning of a *ko*-phrase is defined differently. A *ko*-phrase is a predicate whose meaning is defined as a property of individuals or a property of propositions. An utterance context c includes the speaker, the addressee, the utterance time. These contribute to the interpretations of indexicals, such as *na* ‘I’, *ne* ‘you’, *cikum* ‘now’, etc.

Now we can interpret the structure in (I), assuming that the expressions α , β , and γ can be interpreted in situ:

- III. a. $\llbracket [\gamma\text{-ko } [v \text{ sayngkakha}]] \rrbracket^{c,w,g}$
 = i) $\lambda y \lambda x \exists z [z = y^w \wedge \text{think}(e, x^w, \lambda w' [\llbracket \gamma \rrbracket^{c,w',g}(y^w)])]$
 ii) $\lambda p \lambda x \exists p' [p'(w) \wedge p(w) \wedge p \subset p' \wedge$
 $\text{think}(e, x^w, \lambda w' [\llbracket \gamma \rrbracket^{c,g} \subset p' \wedge \llbracket \gamma \rrbracket^{c,g}(w')])]$
 = $\lambda p \lambda x \exists p' [p(w) \wedge p \subset p' \wedge \text{think}(e, x^w, \lambda w' [\llbracket \gamma \rrbracket^{c,g}(w') \wedge \llbracket \gamma \rrbracket^{c,g} \subset p'])]$
- b. $\llbracket [\beta\text{-}(I)ul \ \gamma\text{-ko } [v \text{ sayngkakha}]] \rrbracket^{c,w,g}$
 = i) $\lambda x \exists z [z = \llbracket \beta \rrbracket^{c,w,g} \wedge \text{think}(e, x^w, \lambda w' [\llbracket \gamma \rrbracket^{c,w',g}(\llbracket \beta \rrbracket^{c,w',g})])]$
 = ii) $\lambda x \exists p' [\llbracket \beta \rrbracket^{c,g}(w) \wedge \llbracket \beta \rrbracket^{c,g} \subset p' \wedge \text{think}(e, x^w, \lambda w' [\llbracket \gamma \rrbracket^{c,g} \subset p' \wedge \llbracket \gamma \rrbracket^{c,g}(w')])]$
- c. $\llbracket [\alpha\text{-ka } [\beta\text{-}(I)ul \ \gamma\text{-ko } [v \text{ sayngkakha}]]] \rrbracket^{c,w,g}$
 = i) $\exists z [z = \llbracket \beta \rrbracket^{c,w,g} \wedge \text{think}(e, \llbracket \alpha \rrbracket^{c,w,g}, \lambda w' [\llbracket \gamma \rrbracket^{c,w',g}(\llbracket \beta \rrbracket^{c,w',g})])]$
 = ii) $\exists p' [\llbracket \beta \rrbracket^{c,g}(w) \wedge \llbracket \beta \rrbracket^{c,g} \subset p' \wedge$
 $\text{think}(e, \llbracket \alpha \rrbracket^{c,w,g}, \lambda w' [\llbracket \gamma \rrbracket^{c,g} \subset p' \wedge \llbracket \gamma \rrbracket^{c,g}(w')])]$

The verb *sayngkakha* ‘think’ takes a *ko*-phrase and then an individual concept or proposition, yielding a property of individual concepts. This is saturated by the subject α .

In (I), γ is a MP, and it denotes a proposition. However, if γ is an interrogative, it denotes a set of propositions and the rest of the interpretations should be different. I will not discuss what they are like. I point out that even if a *ko*-phrase is analyzed as a predicate, we can maintain the standard semantics of MPs in which a declarative denotes a proposition and an interrogative denotes a set of propositions.